LESSON 3.3

MULTIPLYING RATIONAL NUMBERS

LESSON INTRODUCTION

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.

LEARNING TARGETS

- I can fluently multiply rational numbers.

BENCHMARK COHERENCE

BUILDING ON

MA.6.NSO.2.1 Multiply and divide positive multi-digit numbers with decimals to the thousandths, including using a standard algorithm with procedural fluency.

MA.6.NSO.2.2 Extend previous understanding of multiplication and division to compute products and quotients of positive fractions by positive fractions, including mixed numbers, with procedural fluency.

WORKING TOWARD

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

ESSENTIAL QUESTIONS

- How does the process of multiplying integers apply to multiplying rational numbers?

LESSON NARRATIVE

Students will also practice solving multiplication expressions and will determine if the statements will remain true for all rational numbers. Students use a calculator to determine the solutions to these expressions. Then, students will find the product of two rational numbers and will work with a partner to agree on one solution for each of the products. The lesson wraps up with two rational number expressions and three fill-in-the-blank statements about the products of two factors.

COMPONENT SUMMARY

3.3.1 WARM-UP *(~10 MIN.)*

Write and solve integer equations

3.3.2 COLLABORATION *(15-20 MIN.)*

Simplify rational number expressions and complete statements

3.3.3 GUIDED PRACTICE *(10-15 MIN.)*

Find the product of two rational numbers

3.3.4 WRAP-UP *(~5 MIN.)*

Find the product of rational number expressions and complete statements

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Factors

MATERIALS

- Calculator for 3.3.2 Collaboration
- (Optional) Integer chips
- (Optional) Colored pencils

ADDITIONAL PREPARATION

- An anchor chart showing operation rules for rational numbers may be useful.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may assume that multiplying two negative numbers will result in a negative product.
- Students may confuse the different processes to multiply mixed numbers, fractions, decimals, and integers.

THINGS TO CONSIDER

- 3.3.1 Warm-Up may take some students a while to complete, especially if they are using a "guess and check" method.
- Students should only use a calculator for the verification in 3.3.2 Collaboration. Consider how to monitor your students. Options could be that they do not receive a calculator until they can show you, they did the work or to come back as a whole group for the verification.

LITERACY AND LANGUAGE ROUTINES

True or False

Many statements are presented in the 3.3.2 Collaboration. Consider using a True or False routine once students have filled in the blanks to complete the statements. Post statements that have different answers for the blanks and have students justify which statements are true or false.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Mathematical Fluency

Mathematical Fluency requires flexibility and precision while carrying out calculations. By providing students with multiple opportunities to execute procedures and perform calculations, this lesson allows students to maintain flexibility while accurately multiplying rational numbers.

DIFFERENTIATE INSTRUCTION

As students engage with both fractions and decimals throughout this lesson, consider utilizing the On-Ramp to 7h Grade Accelerated Math Learning Tool to provide scaffolding for students in need of additional support, particularly students who learn well with Study Expert videos and practice (available in English and Spanish).

Lesson 3 provides practice opportunities for students to engage in a variety of tools and resources that provide differentiated support. Students can engage with On-Ramp videos on the following topics for additional support:

- Multiplying Decimals
- Multiplying Two Fractions

3.3.1 WARM-UP: BELL WORK

TEACHER MOVES

The Bell Work can inform you about which students understand operations with integers. Although Lessons 1 and 2 were focused on extending operations with integers to rational numbers this open-ended question requires more cognitive demand on student understanding. Encourage students to leave their thinking process as it could help you consider where a student is struggling.

3.3.1 Warm-Up: Bell Work

1. Using the digits 1 to 6 at most one time each, fill in the boxes so that the top two equations are equal, and the bottom equation has the greatest value.

$$-\boxed{6} + \boxed{3} =$$

$$-\boxed{1} - \boxed{2} =$$

$$-\boxed{4} - {}^{-}\boxed{5} =$$

3.3.2 Collaboration: Signed Number Rules

1. Fill in the blank with the words *positive* or *negative* to describe the rules for multiplying integers.
   - The product of two positive integers is a *positive* integer.
   - The product of two negative integers is a *positive* integer.
   - The product of a negative integer and a positive integer is a *negative* integer.
2. Discuss with your partner whether you think the same rules apply to rational numbers. Write a summary of your reasoning.
   Answers will vary. I think the same rules apply because integers are rational numbers.
3. Complete the table. Use a calculator to find the product of each expression to verify if the same rules for signs apply when multiplying rational numbers.

Expression	Sign of Product Using the Rules	Product	Did the Same Rule Apply?
$\left(-\frac{3}{5}\right)\left(-\frac{3}{11}\right)$	+	$\frac{9}{55}$	*yes*
$\frac{7}{18}\left(^{-}4\frac{8}{25}\right)$	-	$-1\frac{17}{25}$	*yes*
$21.285 \cdot {}^{-}3.8$	-	-80.883	*yes*
$({}^{-}91.09)({}^{-}6.4)$	+	582.976	*yes*

4. Complete the sentence.
   The rules for multiplying rational numbers are
   ○ the same as
   ○ different from
   the integer rules for multiplication.

3.3.2 COLLABORATION: SIGNED NUMBER RULES

DIFFERENTIATE INSTRUCTION

Students may not recall the definitions of each vocabulary word in these statements. Provide numerical examples along with each statement so that the students can connect each concept to a clear numerical representation of the statement.

TEACHER MOVES

Lessons 1 and 2 showed students that the rules of adding and subtracting integers applies to rational numbers. Students may base their response on the fact that it worked in the last two lessons Consider asking these students if an integer is considered a rational number. It may also be helpful for students to understand the connection of the types of numbers from whole numbers to rational numbers.

TEACHER MOVES

True or False

Once students have completed the statements from questions 1 and 4, engage students in a True or False routine to synthesize the information presented in this section. Display statements with different answers and have students discuss and explain which statements are true/false and how they know.

3.3.3 GUIDED PRACTICE: APPLYING SIGNED NUMBER RULES TO RATIONAL NUMBERS

TEACHER MOVES

During the Guided Practice, students must find their own answers and then work with their partners to agree on one solution. Prior to completing the table, discuss strategies students can use to help them come to an agreement on one solution.

Students may suggest that each partner solves all questions, and once both partners reach the same answer, they choose that answer as the solution.

DIFFERENTIATE INSTRUCTION

Offer students the option to convert each answer into an equivalent form. Some students may feel more comfortable working with fractions or with decimals, so an opportunity for differentiation is to allow them to convert to the equivalent rational numbers of their choice.

DIFFERENTIATE INSTRUCTION

For students who need additional support with either fractions or decimals, provide On-Ramp video and practice tools to supplement their learning.

QUESTIONING

If time permits, challenge students to determine how many different possible sums they can find and order their approximations from least to greatest.

3.3.3 Guided Practice: Applying Signed Number Rules to Rational Numbers

1. With your partner, decide who will be Partner A and who will be Partner B. Then, find the product of the expression in the column that matches your letter.

Column A	Product	Column B
$790\left(\frac{1}{10}\right)$	79	$7.9(10)$
$\left(-\frac{6}{7}\right)\cdot 7$	-6	$(0.1)\cdot(-60)$
$2.1\cdot(-2)$	-4.2	$(-8.4)\left(\frac{1}{2}\right)$
$2.5\ \cdot\ ^{-}3.25$	-8.125	$\left(-\frac{5}{2}\right)\left(\frac{13}{4}\right)$

The expressions in each row have the same product. Check your work with your partner and, if necessary, resolve any differences. Write the agreed-upon product in the center column.

2. Choose a value from Row A and a value from Row B that, when multiplied, will result in a negative answer. Then, calculate the product.

Row A

$-9\frac{2}{7}$	$-\frac{3}{5}$	$12\frac{2}{9}$

Row B

$4\frac{3}{8}$	$\frac{11}{16}$	$-\frac{9}{50}$

Row A		Row B		Product
$\square$	×	$\square$	=	$\square$
$-9\frac{2}{7}$		$4\frac{3}{8}$		$-40\frac{5}{8}$
$-9\frac{2}{7}$		$\frac{11}{16}$		$-6\frac{43}{112}$
$-\frac{3}{5}$		$4\frac{3}{8}$		$-2\frac{5}{8}$
$-\frac{3}{5}$		$\frac{11}{16}$		$-\frac{33}{80}$
$12\frac{2}{9}$		$-\frac{9}{50}$		-2.2

3.3.4 Wrap-Up: Ticket Out the Door

1. Find the product of each expression.

   a. $(0.24)(^{-}3.8)$
   -0.912

   b. $\left(-\frac{2}{5}\right)\left(-\frac{9}{8}\right)$
   $\frac{9}{20}$

2. Fill in the blanks with the words *positive* or *negative*.
   - Two positive factors result in a *positive* product.
   - Two negative factors result in a *positive* product.
   - A positive factor and a negative factor result in a *negative* product.

3.3.4 WRAP UP: TICKET OUT THE DOOR

QUESTIONING

Allowing the students to solidify their responses with examples of each statement is an opportunity to extend the lesson. Encourage the students to come up with their own numerical equations that would support each statement.

